

CAR BRAIN CHALLENGE
RE-INCARNATE HERBIE

TASK 2

PROTOTYPING-PHASE
BUILD THE INTELLIGENCE

Period: 11 May 2026 – 13 September 2026



Inspired by Herbie the Love Bug (© Walt Disney), this global Challenge invites interdisciplinary university teams to bring the vision of a car as a true personal companion to life.

Introduction to the Challenge

With the Car Brain Challenge, Volkswagen invites students from top universities worldwide to explore future vehicle functions, services, and user experiences in a cloud-native and software-driven environment. The challenge encourages the combination of creative thinking and technical reasoning within a structured, multi-phase process. Each phase builds on the outcomes of the previous one and introduces increasing technical depth and concreteness.

In **Phase 1 (Concept Phase)**, a broad range of innovative ideas was developed based on real-world use cases and future mobility scenarios. From this ideation phase, a Top 10 selection of ideas was submitted per participating group. These submissions were reviewed by Volkswagen with the support of an expert panel and reduced to a Top 3 selection. The shortlisted ideas were subsequently elaborated into fully described concepts. Based on these submissions, one concept per team was selected by Volkswagen to be taken forward into the next phase of the challenge.

Phase 2 (Prototyping Phase) focuses on the technical realization of the selected concept. The concept is translated into a cloud-based prototype in order to explore technical feasibility, system behavior, and implementation challenges in a practical setting.

In the final **Phase 3 (Implementation Phase)**, up to three concepts are further developed with the objective of transferring key aspects of the solution into a vehicle-related environment.

This document describes the tasks, objectives, and framework conditions for Phase 2 of the Car Brain Challenge. It builds on the conceptual work completed in Phase 1 and defines the expectations for the technical prototyping phase, which forms the foundation for all subsequent development steps.

"Our global technology competition calls on the brightest minds from universities around the world to push boundaries and turn this innovative future into reality."

Nikolai Ardey
Executive Director
Volkswagen Group Innovation





Objective of Phase 2

Phase 2 builds on the structured ideation process of Phase 1 and starts with a single, clearly defined concept selected for further development. The objective of this phase is to translate this concept into a **functional cloud-based prototype** and to explore its technical feasibility, behavior, and limitations. This phase focuses on implementation rather than ideation and marks the transition from abstract concepts to a **technically verifiable implementation** that can be analyzed and evaluated.

Teams are expected to realize the core elements of their concept and to make explicit **technical decisions** regarding architecture, interfaces, data flows, and system behavior. The goal is not to deliver a fully mature or optimized product, but to provide a **clear and well-structured implementation** that reflects the intended functionality, technical clarity, and a convincing realization of the underlying idea.

A central objective of Phase 2 is to ensure **transparency, comparability, and verifiability** of the developed prototype. Prototypes must therefore be provided in a form that allows **external execution, inspection, and evaluation** outside the original development environment. This enables assumptions from Phase 1 to be validated, technical trade-offs to be discussed and strengths and limitations of the chosen approach to become visible.

Phase 2 explicitly acknowledges the diversity of concepts in terms of objectives, scope, and technical requirements. **No** uniform technical solution or predefined technology stack is prescribed. Instead, **well-founded implementation choices are expected**, including

- decisions on overall system architecture,
- technologies and frameworks to be used, and
- the appropriate level of technical depth for individual components

Regardless of the approach chosen, the prototypes are expected to:

- go **beyond purely visual or conceptual mock-ups** by implementing essential internal system logic,
- demonstrate **real system behavior**, such as data processing, decision-making logic, or interactions, and
- **clearly demonstrate** how the underlying concept operates in practice.

This does not require all system aspects to be fully implemented. However, core internal components, such as decision-making mechanisms or reactions to inputs, should be implemented wherever feasible rather than simulated. External inputs, such as sensor data or vehicle-related signals, may be mocked in Phase 2 where necessary.

The cloud-based prototype developed in Phase 2 forms the foundation for evaluation and comparison within the challenge and, for selected concepts, for **subsequent technical refinement and transfer into a vehicle-related environment**. Clear structure, comprehensible documentation, and a transparent distinction between the conceptual idea and its technical implementation are therefore essential.

To conclude Phase 2, the developed prototypes are **presented to a jury of experts**. This presentation provides an overview of the current state of the cloud-based prototype and its underlying technical decisions, and enables an in-depth discussion of the chosen approach, its feasibility, and its strengths and limitations within the context of the challenge.



Phase Structure

Time frame:

May 11th to September 13th, 2026

Work period:

All tasks can be completed flexibly within this time frame

Scope of tasks:

- All mandatory tasks must be fully submitted by the end of Phase 2. As the teams' concepts are expected to vary in scope and technical depth, the workload in this phase is intentionally designed to be flexible.
- The mandatory tasks focus on the implementation of a functional, cloud-based prototype that demonstrates the core system behavior of the selected concept. Teams are expected to prioritize technical substance and a coherent implementation over breadth of features.
- In addition to the mandatory requirements, teams are invited to introduce their own ideas and extensions, provided these align with the spirit of the Challenge and meaningfully enhance the technical maturity and expressiveness of the solution.



Submission Format and Deadline

Deadline:

- Cloud-based prototype and documentation:
6 September 2026 23:59 AOE
- Final presentation (PowerPoint):
10 September 2026, 23:59 AOE

Formats:

- The submission format of the cloud-based prototype depends on the developed concept and must be provided in a way that allows an objective evaluation of the solution. This includes all technical artifacts required to execute and assess the prototype (or applicable retrieval mechanisms if said artifacts are publicly available), such as source code, configuration files, and, where applicable, container definitions (for example a Dockerfile).
- The accompanying documentation may be submitted, for example, as a PDF. Other commonly used formats, such as HTML, are also acceptable, provided they can be opened with standard, license-free software.
- The final presentation must be submitted as a PowerPoint file or in a comparable presentation format.
- In addition, supplementary materials are welcome, such as video demos or technical walkthroughs, if they support the understanding of the solution.

Submission:

- All required materials must be submitted by the respective deadlines to the Car Brain Challenge mailbox at car.brain.challenge.vwag.r.wob@volkswagen.de
- If files cannot be sent by email due to size limitations, an upload option will be provided. Access to the upload platform will be granted to the official contact person appointed by the respective university. Uploads must be handled via this contact person.
- Please note that, for data protection reasons, external download links cannot be accepted.
- As part of Task 2, two presentations are required. During the course of the phase, all teams must conduct an intermediate progress presentation by no later than calendar week 27. This presentation is intended to provide an overview of the current status of the technical concept and to enable early content validation and structured feedback by the Car Brain Challenge team. At the end of Phase 2, all teams are required to present their final results in an online presentation including a demonstration of the cloud-based prototype. These final presentations will take place between 7 September and 10 September 2026. Time slots must be booked in coordination with the Car Brain Challenge team. If these dates are not feasible for compelling reasons, teams are required to contact the Car Brain Challenge team at the beginning of Phase 2 to jointly identify a suitable alternative.



Evaluation Criteria

The submissions in Phase 2 will be assessed holistically by an interdisciplinary expert panel. The evaluation focuses on the quality of the technical implementation, the coherence between concept and prototype, and the suitability of the solution as a basis for further technical development and, where applicable, a transfer into a vehicle-related environment. The following criteria guide the evaluation:

- **Technical implementation and functionality**

To what extent has the concept been implemented as a runnable cloud-based prototype? Does the system demonstrate real behavior (for example data processing, decision logic and interactions) and is the solution testable and understandable?

Teams are encouraged to define clear hypotheses or acceptance criteria in advance (for example "If X is triggered, the prototype should demonstrably do Y") and to verify them within the prototype. The evaluation will consider both a convincing demonstration and the system's behavior under demanding or extreme conditions (for example faulty inputs, interruptions and edge cases).

- **Coherence between concept and prototype, including meaningful evolution**

Does the prototype consistently build on and further develop the core idea, user journey and system logic defined in Phase 1? Are the technical decisions clearly derived from the concept, and do they contribute to strengthening and concretizing it in the prototype?

Phase 2 is expected to deepen and refine the Phase 1 concept through implementation. A coherent evolution of the concept is therefore part of the process, as long as the resulting prototype remains logically consistent, well substantiated, and aligned with the original intent. If it becomes apparent during Phase 2 that the selected concept cannot be implemented as initially defined, this must be communicated at an early stage. Any necessary adjustments are to be clarified in coordination with the Car Brain Challenge team.

- **Technical plausibility, feasibility and architecture**

Is the chosen architecture clear, well-structured, and extensible, and does the prototype demonstrate

technical feasibility within the scope of Phase 2?

Are relevant technical constraints, such as latency, data flows, security and privacy, cloud-edge separation, and available compute resources, considered in a sensible way?

The evaluation also considers whether expected failure modes and limitations are identified and appropriately addressed, for example through error handling, fallback mechanisms, or a transparent discussion of known constraints.

- **Maturity of implementation**

How far has the system been developed technically?

Both highly innovative approaches with limited functional scope and less novel, but more thoroughly implemented systems can be rated positively, as long as they are technically convincing and well justified.

- **Quality of documentation and presentation**

Is the solution clearly documented and explained? Do the codebase, documentation and presentation enable a well-founded evaluation and discussion?

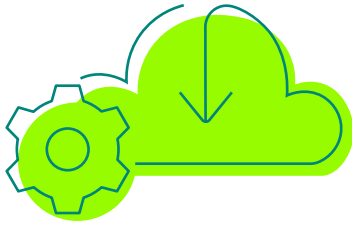
At the end of Phase 2, the results will be presented to an expert team. Documentation and communication should therefore provide a comprehensive overview of the current state of the cloud-based prototype, justify key decisions, and transparently state assumptions, simplifications and open points.

- **Efficiency and resource awareness**

How efficient is the solution in terms of compute usage, runtime, memory consumption, and overall system complexity? Is it evident that technical choices have been made with later feasibility and operability in mind, for example with respect to edge capabilities or constraints in vehicle-related environments?

The evaluation focuses on architectural and design decisions rather than model optimization. Fine-tuning of models is not required in Phase 2. Instead, an awareness of resource implications and scalability considerations is expected at a conceptual and architectural level.

The evaluation does not focus on the use of specific technologies, programming languages or frameworks, but on the quality, substance and overall credibility of the solution.



Technical framework

For Phase 2, the Car Brain Challenge gives teams **full freedom to choose the cloud infrastructure** they use for the implementation of their cloud-based prototype.

Universities may use their own cloud environments, university computing resources, or external cloud services, provided that the prototype is functional, transparently documented, and testable by Volkswagen in a comprehensible manner.

Alternatively, Volkswagen offers teams the option to use a **cloud development environment within the Microsoft Cloud of Volkswagen**. This offer is intended as a **support option** and is **not mandatory**.

To access the Volkswagen cloud environment, each participating individual requires a **personal Volkswagen user account**. For reasons of **IT security and compliance**, such user accounts can only be provided if the respective university is registered as an **official partner of the Volkswagen Group**.

The relevant **guidelines and organizational requirements** have already been shared with the universities separately via email and are required for access to the Volkswagen cloud environment.

For the implementation of the prototypes in Phase 2, Volkswagen provides a **budget of 6,000 € per participating university**.

The technical framework conditions of Phase 2 are intended to ensure that:

- all teams can implement their concepts in a functional and verifiable way,
- different technical approaches remain possible,
- and the results are presented in a way that enables fair evaluation and constructive discussion.

The focus is **not on the full utilization of all provided resources**, but on a **meaningful implementation aligned with the respective concept**.

It is **neither required nor expected** to build a **full digital twin of a Volkswagen vehicle**. The focus of Phase 2 lies on prototypical system behavior, interaction logic, and technical feasibility, not on a complete or vehicle-accurate simulation.

Container-Based Execution

To ensure reproducible and device independent execution of the prototypes, a **container-based deployment** (e.g. Docker) is **explicitly recommended**. Containers offer the advantage that applications, including their dependencies, can be executed consistently across different systems.

The use of containers is **not mandatory**. Teams may choose alternative, comparable solutions, provided that they:

- are based on **open source technologies**,
- allow for **transparent and reproducible execution**, and
- make it possible, with reasonable effort, to **test and evaluate the prototype system agnostically outside the team's own development environment**.

Regardless of the chosen approach, teams are expected to provide **transparent access to the underlying implementation** (e.g. source code, configurations, build artifacts) to enable proper evaluation of the prototype.



Technical Onboarding and Support

As Phase 2 involves a significantly increased level of technical depth, comprehensive support is provided to facilitate the participating teams throughout this phase. At the beginning of Phase 2, a technical documentation package is made available, complemented by a live technical onboarding session to address specific questions and clarify expectations. In addition, individual support is provided throughout Phase 2 to assist with particularly challenging technical issues and to support progress during the implementation phase.

Onboarding

In addition to this task, one individual technical onboarding session is conducted per university. During this session, open questions regarding the Phase 2 task are addressed and the planned cloud setup is discussed, depending on which cloud solution the respective team chooses.

If a university decides to use its own cloud infrastructure, the onboarding session primarily serves as a technical exchange between the university team and Volkswagen. The objective is to ensure early on that the developed prototype can later be operated within the Volkswagen cloud environment.

If a team opts to use the Volkswagen cloud, the onboarding session focuses on explaining the relevant technical framework conditions, requirements, and constraints to the students, as well as clarifying any open questions.

Support

Support is provided on an individual basis for each team and can take place at regular intervals or as needed. Teams are encouraged to proactively approach the organizers to request support and to coordinate or schedule support sessions independently.

Task Phase 2

In Phase 1, the three shortlisted concepts were further elaborated and prepared for continued development. Based on the evaluation by the jury, each participating group received a recommendation indicating which concept should be taken forward into Phase 2.

In Phase 2, the selected concept is to be translated into a **substantial and functional cloud-based prototype**. Task 2 defines the concrete activities and deliverables to be completed during this phase.

The work in Phase 2 is structured around the following three main components:

- **Implementation:** development of the cloud-based prototype
- **Documentation:** description of the implemented solution and its underlying assumptions and constraints
- **Presentation:** presentation of the final Phase 2 solution

→ 1. Implementation

The objective of the implementation is to make the conceptual ideas developed in Phase 1 technically tangible and to **demonstrate how the system operates in practice**. The implementation is deliberately **technology-agnostic**. **No specific** programming languages, frameworks, or AI models are prescribed. What matters is the **functional substance** of the prototype and the **coherence between the underlying concept and its technical realization**.

Both highly innovative, experimental solutions and technically advanced systems can be successful, provided that they are convincingly implemented as a **working prototype**. A **runnable cloud-based prototype** is required that represents the core system behavior of the concept.

The prototype is expected to:

- be **executable**,
- be **testable**, meaning that core functionalities can be traced and verified,
- demonstrate **real system behavior**, including data processing and decision-making, and
- produce **outputs** such as reactions, state changes, or feedback.

→ 2. Documentation

Teams are required to provide a comprehensive qualitative documentation of the technical elaboration of their solution. The documentation should be broadly structured along the following chapters:

- Technical Architecture and System Structure
- Context and Data Model
- Technical Implementation and Artefacts
- Contextualization and Further Development

If additional information beyond these chapters is considered useful for a thorough assessment and informed discussion of the solution, it may be included in the documentation.

The focus of the documentation lies on **clarity and traceability** rather than formal completeness. As the concepts may differ significantly in scope and technical approach between teams, the exact structure and level of detail of the documentation can be defined by the teams themselves, provided that the solution remains comprehensible and well substantiated.

2.1 Technical Architecture and System Structure

A **clearly defined technical architecture** must be designed and implemented for the prototype.

This includes:

- a description of the **main system components**, such as services, modules, or agents,
- an illustration of **data and control flows** between these components, and
- a rationale for **key technical design decisions**.

The architecture should demonstrate:

- how the system is structured in a modular way, and
- which responsibilities are assigned to which components.

A fully production-ready architecture is **not** expected. However, the proposed structure should be **coherent, technically plausible**.

In real-world applications, components-based on large language models are typically fine-tuned on domain-specific data to improve performance and behavior. Due to the associated effort, such fine-tuning is not required in Phase 2. However, in preparation for a potential Phase 3, the technical documentation should indicate which components would benefit from fine-tuning and explain the rationale behind these considerations.

2.2 Context and Data Model

A **context and data model** must be developed to describe how contextual information is represented and used within the system.

The expected outcome includes:

- a structured representation of the relevant **context dimensions**, such as user, vehicle, environment, system, or interaction context,
- a description of the **relationships between these context dimensions**, and
- an explanation of how contextual information is **processed, stored, and propagated within the system**.

Within the scope of this challenge, the addressed use cases typically cover only a subset of the overall system capabilities. It is therefore important to design the context model in a **clean and generic** way that allows for **future extension**. The technical concept should outline how the system can be expanded to incorporate additional context sources, use cases, or models over time.

The objective is **not to create a formally complete or academically rigorous ontology**, but to provide a **clear and comprehensible model** that demonstrates how context serves as a foundation for intelligent system behavior and supports extensibility beyond the scope of the current prototype.

2.3 Technical Implementation and Artefacts

Teams must provide all technical artefacts required to understand, execute and evaluate the virtual prototype.

This includes, in particular:

- the complete source code of the solution,
- configuration and startup files, which must be clearly documented and accompanied by concise instructions on how to run the prototype,
- information on dependencies and build processes,
- and, where applicable, container definitions (for example Dockerfiles).

2.4 Contextualization and Further Development

In addition to the technical documentation of the cloud-based prototype, teams are required to submit a structured contextualization of their solution. This section should enable a well-founded discussion of the prototype and outline potential directions for future development.

As part of this contextualization, teams are expected to:

- describe key assumptions, simplifications and open points underlying the current implementation,
- outline the technical strengths and limitations of the solution,
- explain how the system could be scaled, extended or further developed in principle,
- provide background reasoning and guiding ideas behind the concept,
- outline the underlying design assumptions and premises that guided the technical and conceptual decisions
- and, where applicable, show how the concept from Phase 1 has been further developed, for example by refining the user journey or extending the original concept logic.

The focus of this section is not on completeness, but on clarity, reflection and technical reasoning, providing a transparent basis for discussion and evaluation.



3. Presentation

During Phase 2, two presentations are scheduled: an intermediate progress presentation during the course of the phase and a final presentation at the end of Phase 2.

The intermediate progress presentation is mandatory and must be scheduled by the teams during Phase 2. The teams are required to independently integrate this presentation into their roadmap and coordinate the date at an early stage with the Car Brain Challenge team. The presentation must take place no later than calendar week 27. During this session, the teams present their technical concept.

This intermediate presentation serves solely for content validation and a structured feedback loop provided by us and does not aim to prescribe or influence architectural or technology decisions.

The purpose of this session is to present the current state of work; teams are not expected to prepare an extensive or elaborate presentation.

With this presentation, we aim to support the teams in maintaining the intended scope of Phase 2 and to provide constructive and relevant feedback.

At the end of Phase 2, teams are required to present their solution in a presentation including a demonstration of the cloud-based prototype. The presentation will take place between 7 September and 10 September 2026. The exact time slot will be scheduled individually in coordination with the Car Brain Challenge team. If these dates do not fit at all for compelling reasons, teams are required to reach out to the Car Brain Challenge team at the beginning of Phase 2 to jointly identify a suitable alternative.

Each presentation slot is scheduled **for one hour in total**. The **presentation** of the prototype itself **should not exceed 20 minutes**. This is followed by a live or video-based demonstration showing the cloud-based prototype in action, as well as a discussion and Q&A session.

The primary objective of the presentation is to understand the teams' reasoning and decision-making throughout the development process. The presentation should therefore cover the following aspects:

- an overview of the current state of the cloud-based prototype,
- a brief introduction and contextualization of the concept, including its strengths, limitations, and potential for improvement or scaling,
- the approach taken during the development of the cloud-based prototype,
- an explanation of the technical architecture, including the rationale behind key technical design decisions.

Administration

Conditions of Participation

Participation in the Re-InCARnate Herbie Tech Challenge is governed by a formal contract between the Volkswagen Group, as the organizing entity, and each participating team.

All participating universities are required to review and sign the Conditions of Participation before starting any work related to the Challenge.

Team structure

The composition and structure of the teams are determined by the participating universities. We welcome teams with diverse academic and professional backgrounds to encourage creative and interdisciplinary thinking.

Teams are free to organize and coordinate their internal processes independently.

Each university must appoint an official representative who can support the team and serve as the contact person for the Volkswagen Group.

The Car Brain Challenge Organizer

The Challenge is organized by:

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